

BIKE SALES ANALYSIS PROJECT REPORT

By

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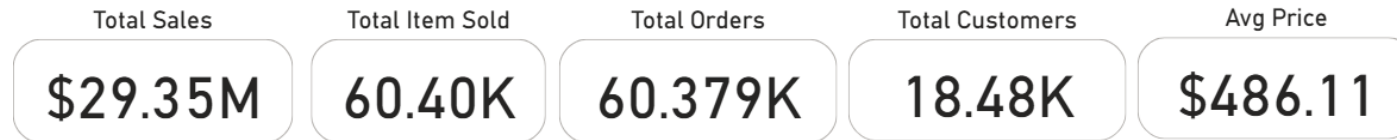
Bike Sales analysis Project

- Project Repository
- Data Warehouse Project Repository
- Datasets

Exploratory Data Analysis

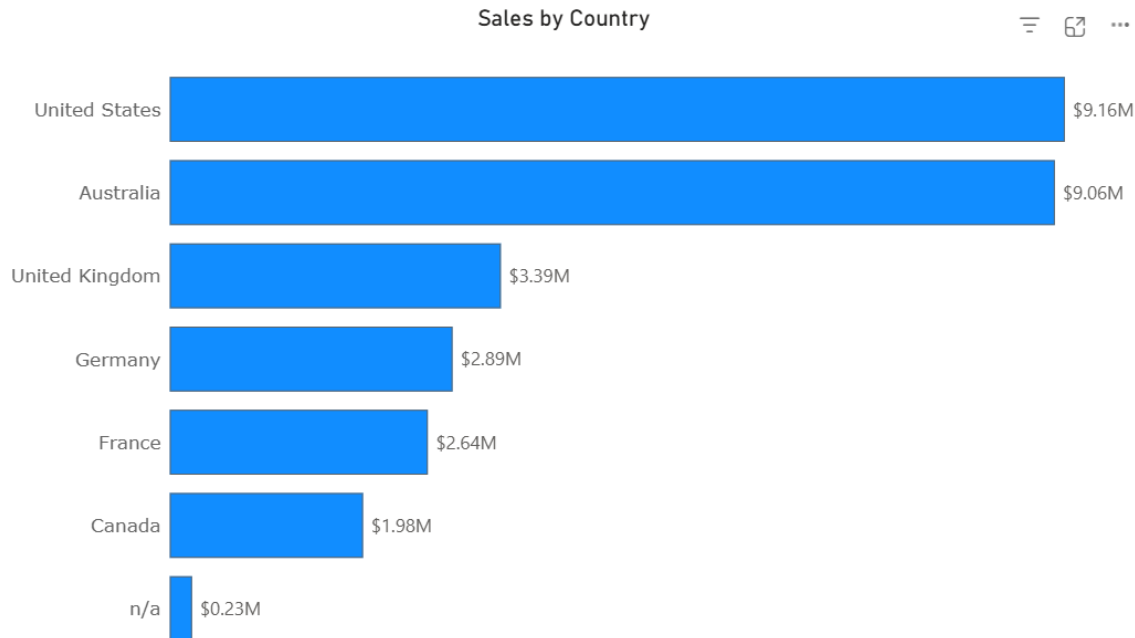
Date range of the data: "2010-12-29" to "2014-01-28"

Key Performance Indicators (KPIs)



- **Key Insights:**
- The business generated **\$29.35 million** in revenue, with over **60,420 products sold**.
- A total of **60,379 transactions** were recorded, placed by approximately **18,484 unique customers**.
- The **average price per item** across all sales is **\$486.11**, indicating a mid-to-premium product range.

Geographic Sales Distribution:



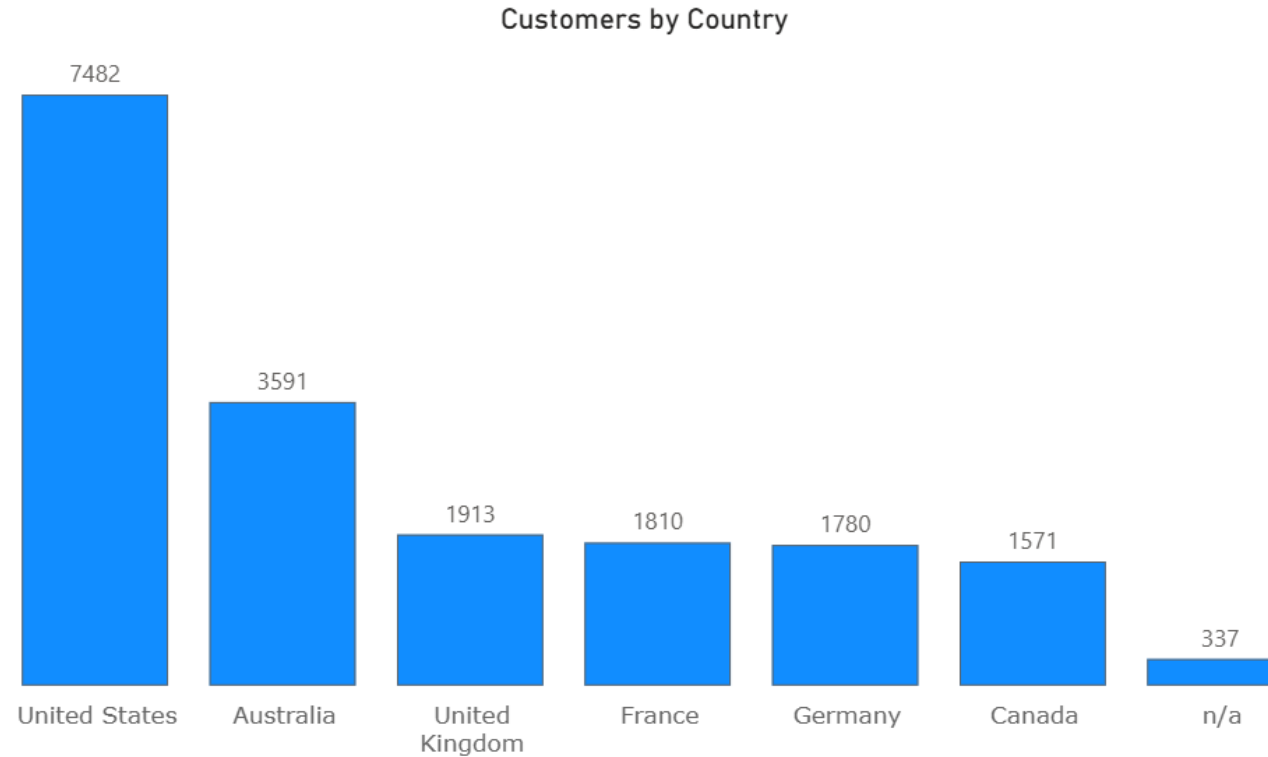
- The **United States (\$9.16M)** and **Australia (\$9.06M)** are the top-performing markets, together contributing over **60%** of total sales.
- The **United Kingdom (\$3.39M)**, **Germany (\$2.89M)**, and **France (\$2.64M)** show strong mid-tier performance.
- **Canada (\$1.98M)** presents potential for growth.
- Around **\$230K** worth of sales have no country associated, indicating data quality issues or unclassified records.
- ***Recommendation:*** Prioritize marketing efforts in the US and Australia, and investigate the “n/a” country data for cleanup.

Items Sold by Country

country	Sum of quantity
United States	20,474
Australia	13,344
Canada	7,629
United Kingdom	6,908
Germany	5,626
France	5,552
n/a	871
Total	60,404

- **United States** leads not only in revenue but also in volume, with **20,474 items sold**.
- **Australia** follows with **13,344 units**, indicating a strong market presence.
- Interestingly, **Canada** ranks 3rd in volume (7,629 items) but 6th in revenue — suggesting **lower-priced items** or **high discount rates**.
- **Germany, France, and the UK** all have similar unit volumes between 5.5K–6.9K, reflecting consistent demand across Europe.
- The **“n/a” category** again highlights data needing attention — **871 units** lack country info.

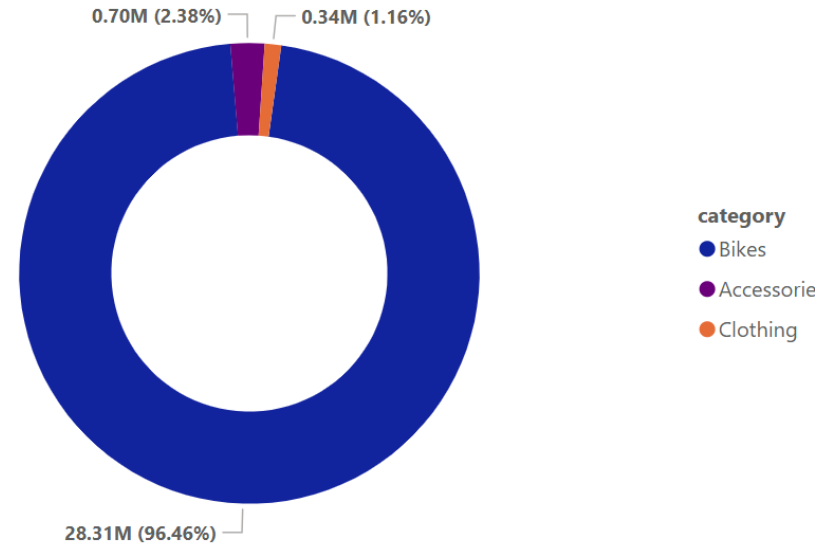
Customer Distribution Insight



- **United States** leads with **7,482 customers** (~40% of total).
- Followed by **Australia** with **3,591**, and the **UK**, **France**, and **Germany** each with over 1,700 customers.
- Other countries like **Canada** (1,571) and the "n/a" group (337) trail behind.

Sales by Category & Subcategory

category	Sum of sales_amount
☐ Bikes	28311657
☐ Accessories	699909
☐ Clothing	339692
Total	29351258



- **Bikes** are the clear leader, accounting for **\$28.31M** in revenue. **96.46%** of total sales.
- **Accessories** and **Clothing** contribute **\$699.9K** and **\$339.7K**, respectively, suggesting potential for growth through bundling or promotions.

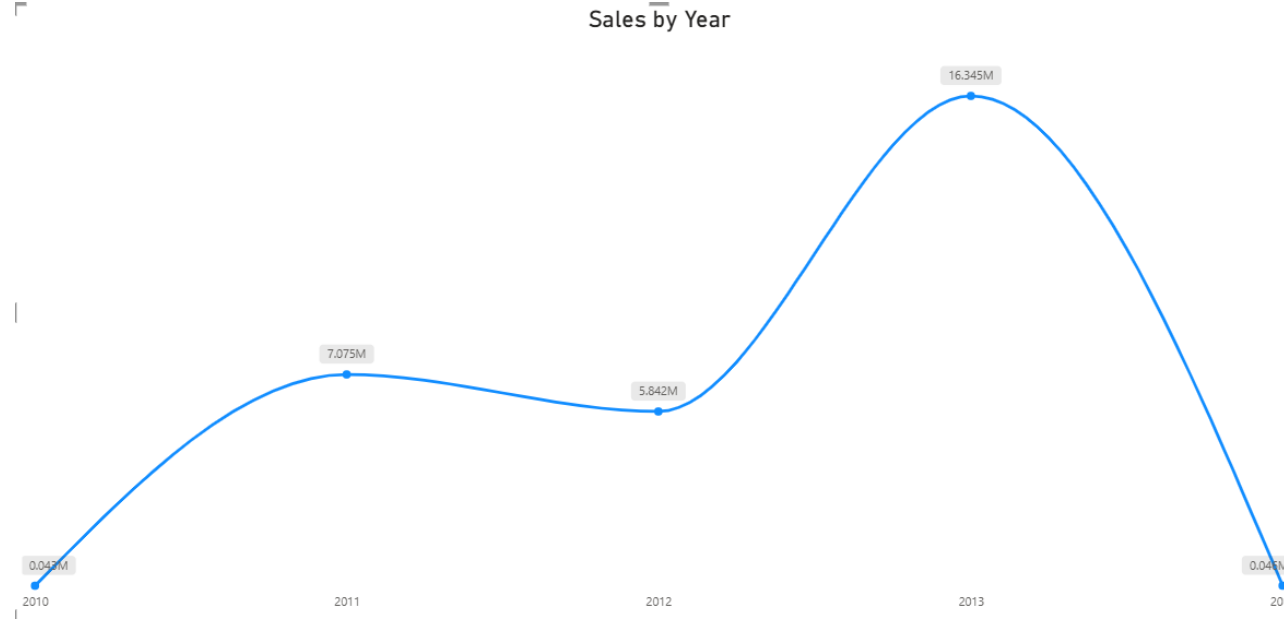
category	Sum of sales_amount
[-] Bikes	28311657
+ Road Bikes	14519438
+ Mountain Bikes	9947639
+ Touring Bikes	3844580

[-] Accessories	699909
+ Tires and Tubes	244445
+ Helmets	225400
+ Bottles and Cages	56984
+ Fenders	46662
+ Hydration Packs	40315
+ Bike Stands	39591
+ Bike Racks	39240
+ Cleaners	7272

[-] Clothing	339692
+ Jerseys	173084
+ Shorts	71330
+ Vests	36160
+ Gloves	34296
+ Caps	19710
+ Socks	5112

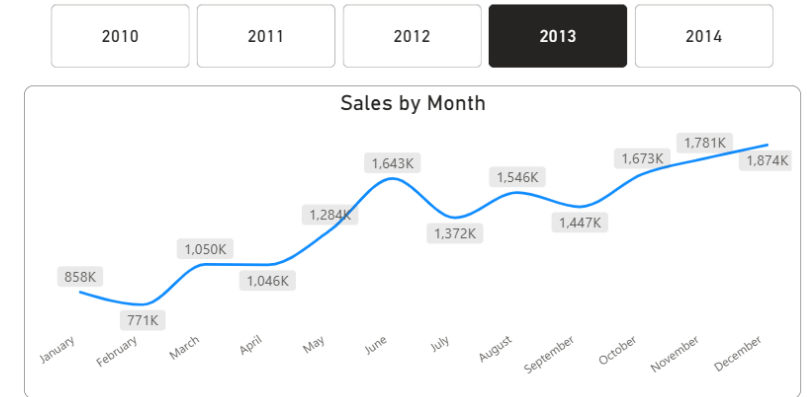
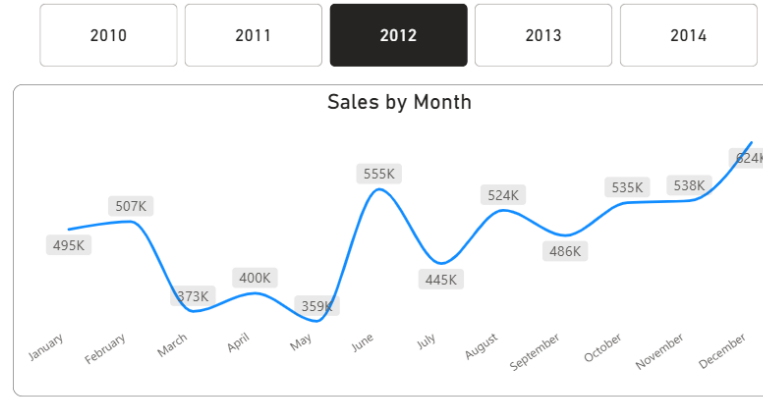
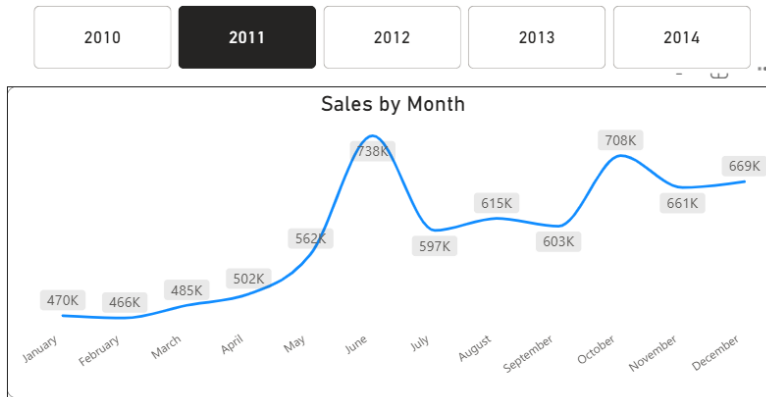
- **Top Performing Subcategories:**
- Within **Bikes**, the best sellers are:
 - **Road Bikes** – \$14.52M
 - **Mountain Bikes** – \$9.95M
 - **Touring Bikes** – \$3.84M
- Within **Accessories**:
 - **Tires and Tubes** – \$244.4K
 - **Helmets** – \$225.4K
 - Others (e.g., Hydration Packs, Bike Stands) are below \$60K
- Within **Clothing**:
 - **Jerseys** dominate with \$173K
 - Followed by **Shorts** (\$71K) and **Vests/Gloves** (~\$35K each)

Sales Trend by Year (2010–2014)



- Sales began in **2010** at **\$43K**, marking the start of operations.
- In **2011**, sales jumped to **\$7.08M**, showing strong early growth.
- In **2012**, there was a slight dip to **\$5.84M**, possibly due to market saturation or seasonal changes.
- A major spike occurred in **2013**, reaching **\$16.35M**, the highest on record — more than **2.5x** growth from 2012.
- **2014** shows only **\$45K** in sales, indicating the year was either incomplete in the data or represents a shutdown/transition period.

Sales by Month



- **Overall Growth:** Sales increased annually, from 470K (Jan 2011) to 1,874K (Dec 2013), showing a strong upward trend.
- **Seasonal Peaks:** June and December consistently saw the highest sales (e.g., 738K in Jun 2011, 1,874K in Dec 2013), likely due to seasonal demand or promotions.

- February and March often had dips (e.g., 373K in Mar 2012), possibly from post-holiday slumps.
- Sales fluctuated but recovered well after dips (e.g., 738K in Jun 2011 to 597K in Jul, then 708K in Sep).
- March 2012 dropped to 373K, while Dec 2013 surged to 1,874K (127% increase from Dec 2012).

Top & Bottom Performing Products

product_name	Sum of sales_amount
Mountain-200 Black- 46	1373454
Mountain-200 Black- 42	1363128
Mountain-200 Silver- 38	1339394
Mountain-200 Silver- 46	1298709
Mountain-200 Black- 38	1292559
Total	6667244

product_name	Sum of sales_amount
Touring Tire Tube	7435
Bike Wash - Dissolver	7272
Patch Kit/8 Patches	6378
Racing Socks- M	2682
Racing Socks- L	2430
Total	26197

Top Performing Products (by Sales Amount):

- All top 5 products are **Mountain-200** bike variants (Black/Silver, sizes 38–46)
- Combined, they generated over **\$6.66M** in revenue
- This suggests **strong demand for high-end mountain bikes**, especially size 42–46

Bottom Performing Products:

- The lowest performers include **Touring Tire Tubes, Bike Wash, Patch Kits, and Racing Socks**
- Together, they contributed less than **\$26.2K**, indicating low traction or poor visibility

Recommendations Summary

Based on our analysis of sales, products, customers, and time trends, here are our key recommendations to drive growth and operational efficiency:

1. Diversify Product Portfolio to Reduce Risk

- While the **Bikes category contributes 96% of total sales**, this over-reliance poses a **significant business risk**. If market demand shifts, supply issues arise, or competition increases, the impact could be severe.
- Invest in growing **Accessories and Clothing** through product innovation, better positioning, and targeted promotions.
- Explore **customer bundle behavior** to understand what complementary products can be cross-sold or upgraded.
- Launch seasonal or trend-based collections to stimulate demand outside core bike sales.
- Diversifying revenue streams will **stabilize the business** and create more **resilience** in the long term.

2. Prioritize Top Markets

- The **United States and Australia** contribute **>60%** of total revenue.
- Increase targeted advertising, partnerships, and stock availability in these regions.

3. Improve Data Quality

- Resolve "**n/a**" **entries** in country and product fields (~\$230K in sales, 871 items).
- Clean, complete data improves analytics, targeting, and reporting accuracy.

4. Plan for Seasonal Peaks

- Sales peaked massively in **2013 (\$16.35M)** after a dip in 2012.
- Investigate what drove this growth (e.g., campaigns, launches) and replicate the strategy during future seasons.

5. Boost Cross-Selling Opportunities

- Low accessory and clothing sales suggest missed cross-sell potential.
- Promote **kits, bundles, or checkout suggestions** to raise accessory sales per order.